Hi I am selling fish and i want to know how many fishes to order each day, i will have to order 2 days before selling for the fishes to arrive, the selling will last for 3 months.
The demand is deterministic. The fishes are treated as a perishable product, it expires 3 days after arrival. The holding cost per unit is $2 The lost-sales penalty cost is $3, unsatisfied demand is backlogged. The setup cost is $10 The salvage cost is $0 The unit cost is $25 The time horizon is exactly 3 months

Inventory Model Summary for Fish Sales
1. Time Horizon
- **Duration**: 3 months (~90 days).
2. Demand Characteristics
- **Type**: **Deterministic** (exact daily demand must be specified; currently missing).
- ❗ **Required**: Please provide the **daily demand quantity** (e.g., 100 fish/day).
3. Product Characteristics
- **Perishability**: Expires **3 days after arrival**.
- **Lead Time**: **2 days** (order placed 2 days before selling).
- **Implication**:
- Fish arrive on **Day 2** and must be sold by **Day 5** (Day 2 + 3-day shelf life).
- Since orders are placed daily, this creates overlapping inventory windows.
4. Inventory Policy
- **Unsatisfied Demand**: **Backlogged** (unmet demand accumulates).
- **Order Frequency**: Daily.
5. Cost Structure

Cost Type	**Value**
Unit Cost	**$25/fish**
Holding Cost	**$2/fish/day** (for fish not sold before expiry)
Lost-Sales Penalty	**$3/fish** (backlog cost per unit)
Setup Cost	**$10/order**
Salvage Value	**$0** (expired fish are worthless)

Missing Information
1. **Daily Demand**: Since demand is deterministic, the exact number of fish demanded each day is required (e.g., 150 fish/day).
Key Observations
- **Perishability + Lead Time Interaction**:
- Fish ordered on **Day *t*** arrive on **Day *t+2*** and expire by **Day *t+5***.
- Daily ordering means inventory from multiple orders may overlap.
- **Backlogging**: Unmet demand carries over but incurs a $3/fish penalty.

Next Steps
Once you provide the **daily demand quantity**, the model will be complete. For deterministic demand, the optimal order quantity typically matches demand precisely (adjusted for lead time and perishability).
Please specify:
- What is the **exact daily demand** for fish?
After receiving this, I'll confirm the full model.

the exact demand is 200 fish/day

Let me know if you need further assistance in formulating this model! Hi I am selling fish and i want to know how many fishes to order each day, i will have to order 2 days before selling for the fishes to arrive, the selling will last for …

Base

the demand should follow a uniform distribution from 50 to 200 fish each day but is discrete

Epoch 1

daily demand is 200 fish/day

Epoch 2

the exact demand is 200 fish/day

Epoch 3